Magnetism

Part 2
(Lorentz Force & Electric Motor)

Trains in many cities use electric motors to transport millions of people to and from work each day. In the first part of this chapter, we are going to look at how an electric motor works. An electric motor creates motion from an electric current.

An Electric Motor CREATES MOTION from an electric current

LEARNING OBJECTIVES

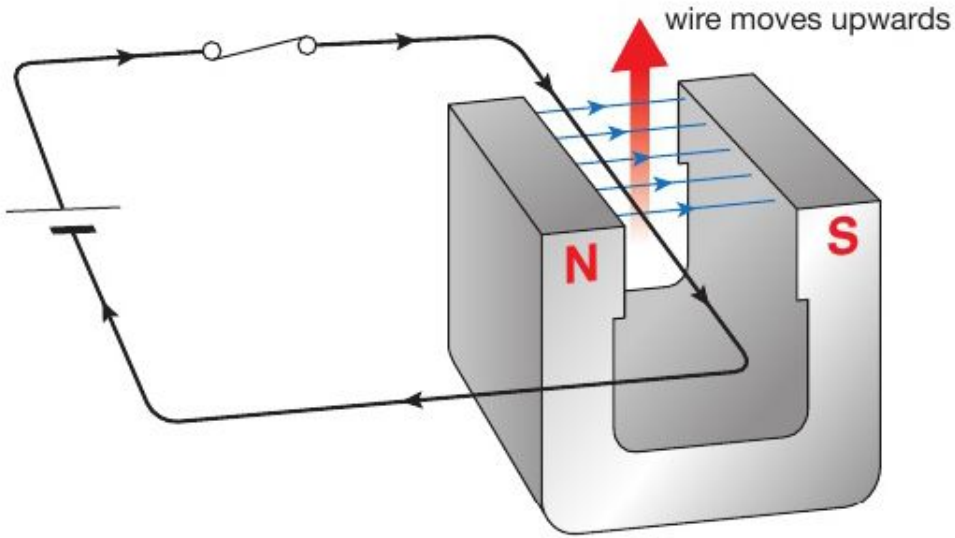
PHYSICS ONLY

- Know that there is a force on a charged particle when it moves in a magnetic field as long as its motion is not parallel to the field

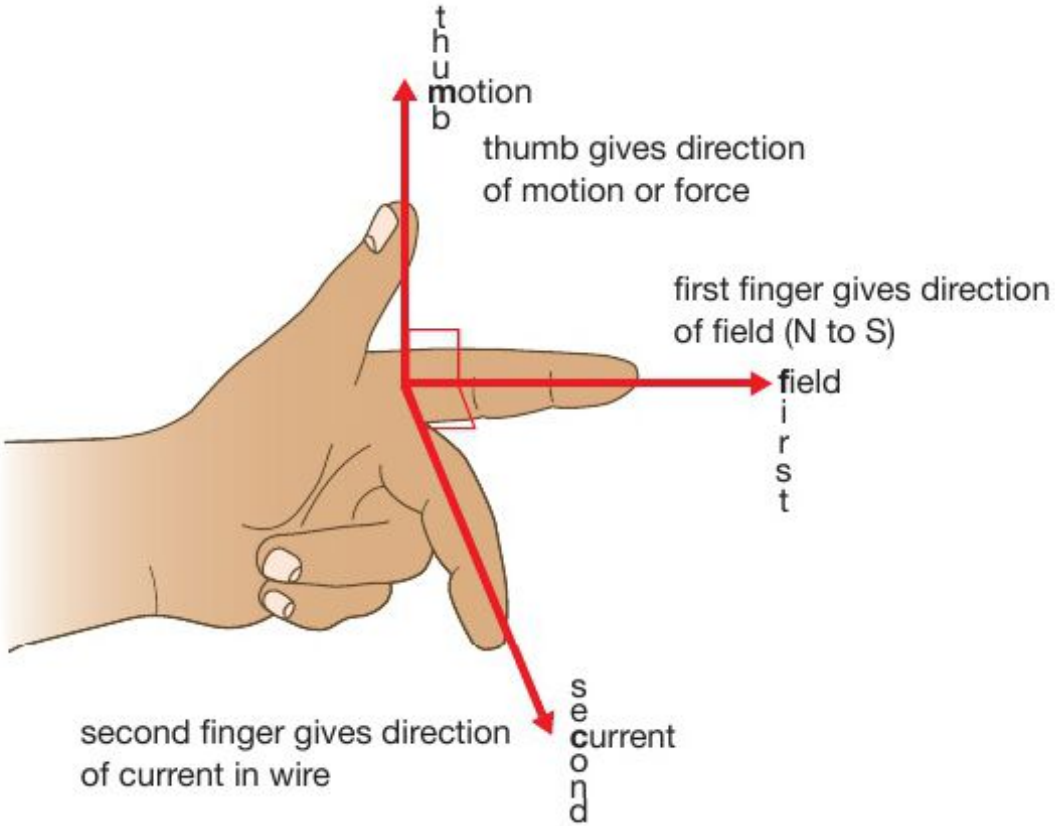
- Understand why a force is exerted on a current-carrying wire in a magnetic field, and how this effect is applied in simple d.c. electric motors and loudspeakers
- Use the left-hand rule to predict the direction of the resulting force when a wire carries a current perpendicular to a magnetic field
- Describe how the force on a current-carrying conductor in a magnetic field changes with the magnitude and direction of the field and current

MOVEMENT FROM ELECTRICITY

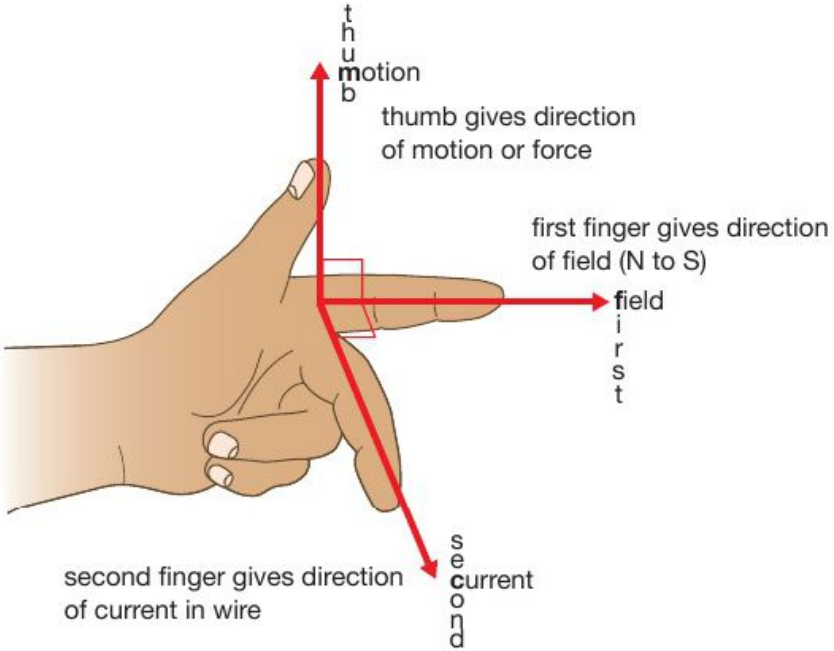
When a charged particle moves through a magnetic field it experiences a force, as long as its motion is not parallel to the field. We can demonstrate this effect by passing electrons (that is, an electric current) along a wire that is placed in a magnetic field as shown in Figure 21.2. When the switch is closed and electrons flow, the wire will try to move upwards.

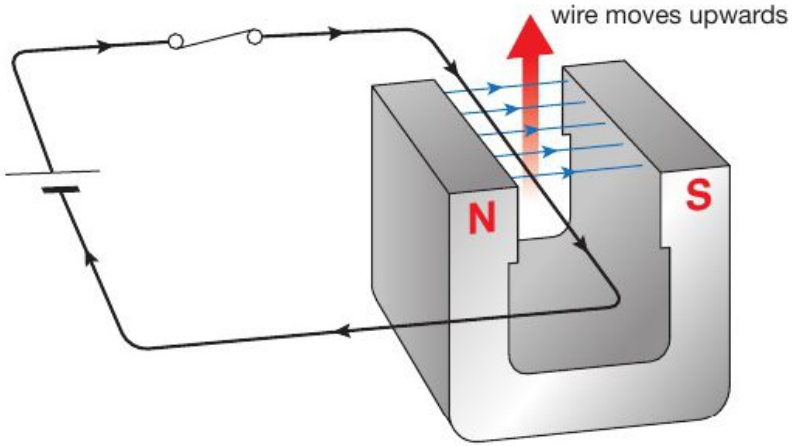


▲ Figure 21.2 The wire moves where there is a current in it.



▲ Figure 21.5 Fleming's left-hand rule helps you to work out the direction of the force.

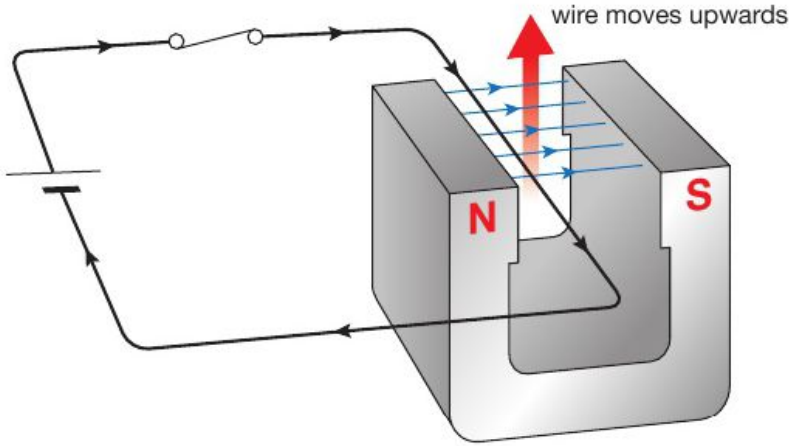




The wire moves where there is a current in it.

A stronger force will be produced if the magnetic field is stronger or if the current is increased.

If we change the direction of the current or the magnet's field, a different overlapping pattern is created and we will see the wire move in the opposite direction.

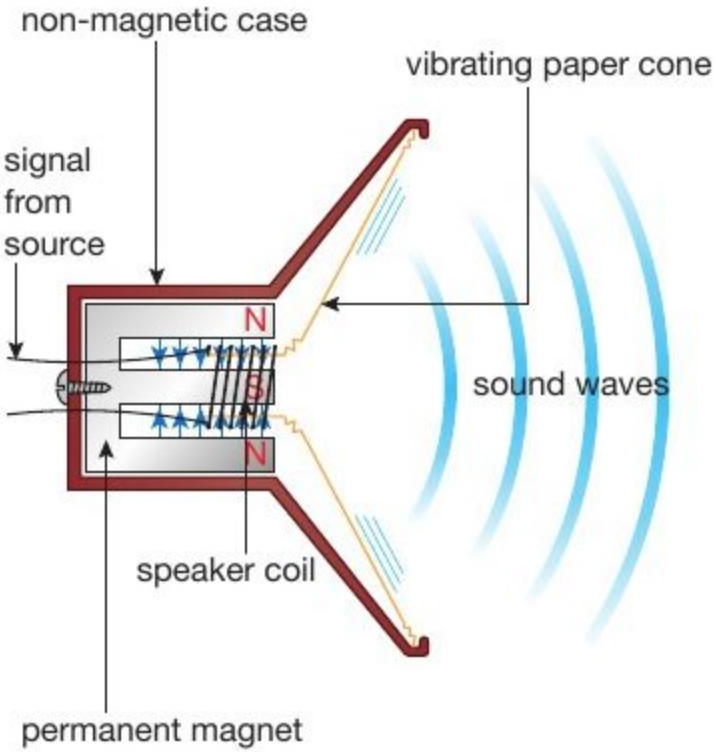


The wire moves where there is a current in it.

THE MOVING-COIL LOUDSPEAKER

The moving-coil loudspeaker uses the motor effect to transfer electrical energy to sound energy.

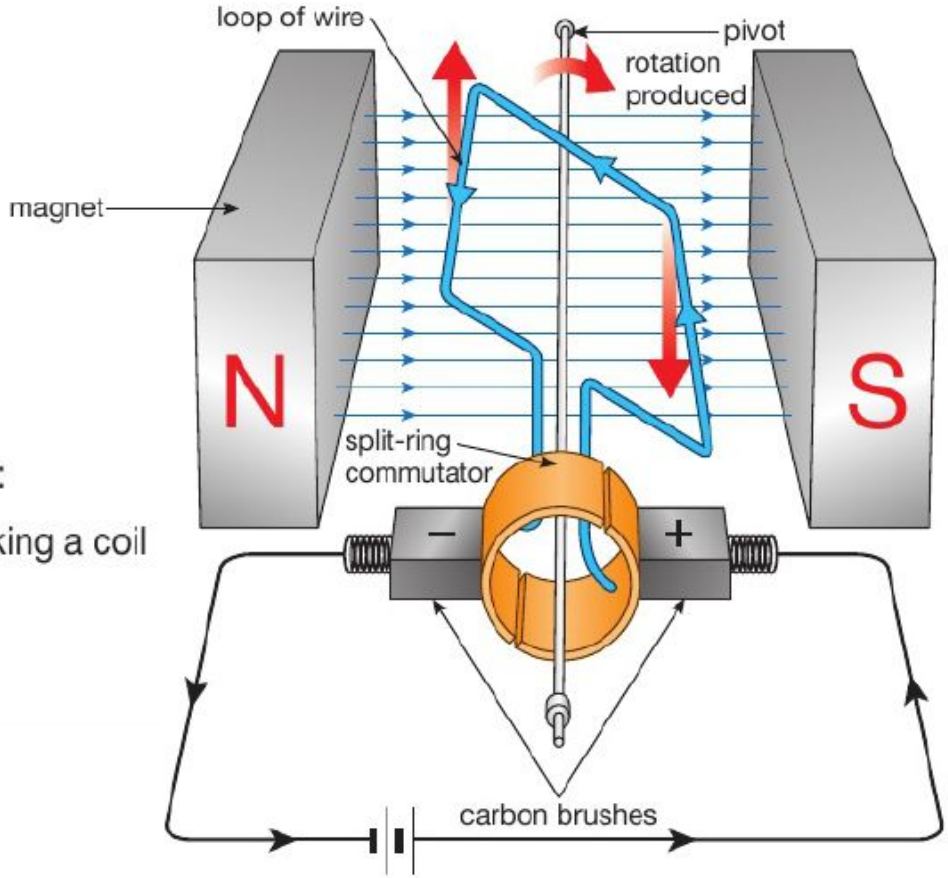
- Electric currents from a source, such as a radio, pass through the coils of a speaker.
- These currents, which represent sounds, are always changing in size and direction, like vibrating sound waves.
- The fields of the coil and the permanent magnet are therefore creating magnetic field patterns which are also always changing in strength and direction.
- These fields in turn apply rapidly changing forces to the wires of the coil, which cause the speaker cone to vibrate.
- These vibrations create the sound waves we hear.



▲ Figure 21.6 A loudspeaker transfers electrical energy to sound energy.

THE ELECTRIC MOTOR

The electric motor is one of the most important uses of the motor effect. Figure 21.7 shows the most important features of a simple d.c. electric motor.



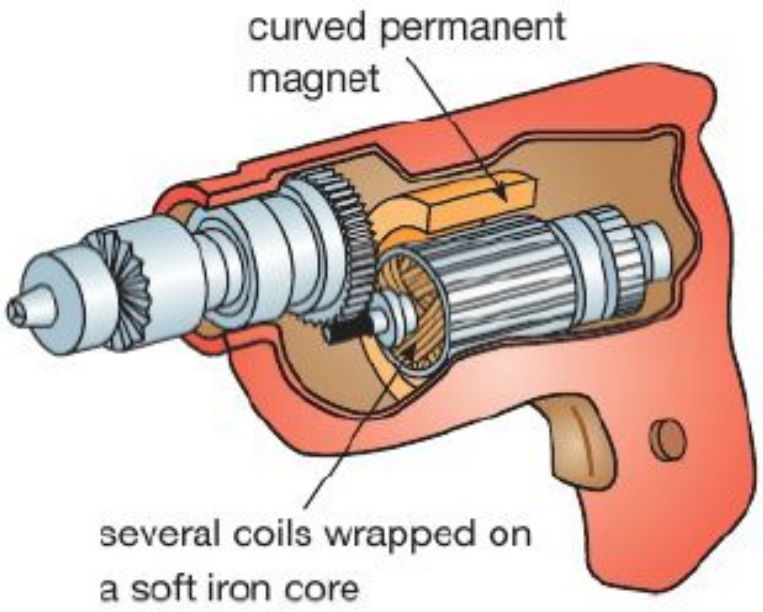
▲ Figure 21.7 A simple electric motor

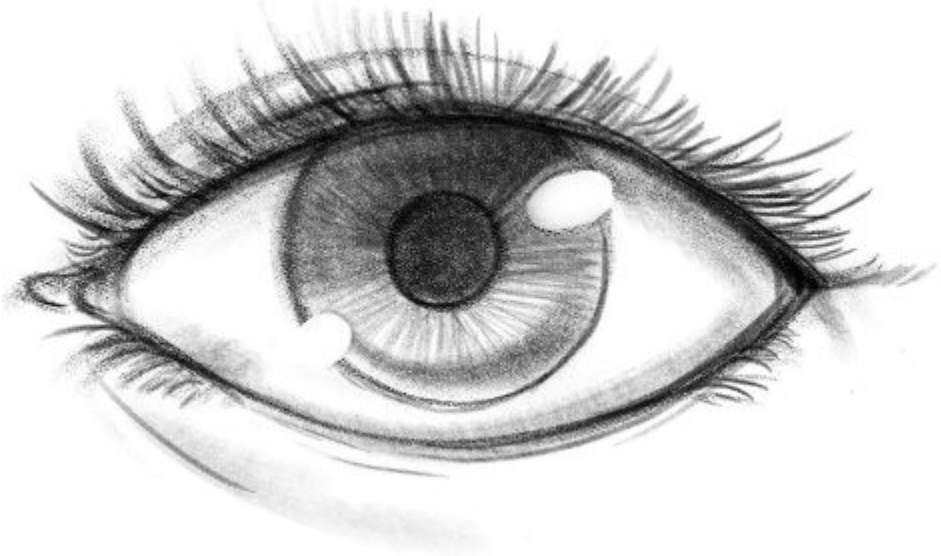
To increase the rate at which the motor turns we can:

1. increase the number of turns or loops of wire, making a coil
2. increase the strength of the magnetic field
3. increase the current in the loop of wire.

APPLICATION:

Any more application??



▲ Figure 21.8 A real electric motor

Good Luck & Happy Reading